

Mechanical Engineering

Name : Aditya Tripathi

UNI Roll no: 1251000116

Class: AF-1

Class Roll No: 3

ques-1)

A) Isochoric process $\Rightarrow$ A process in which vol remains constant ($V = \text{constant}$).

$$W = \int P dV = 0$$

Since $dV = 0$
work done = 0

ex: - heating gas in a rigid tank

B) Isobaric process $\Rightarrow$ (constant pressure process)
A process in which remain constant

work done $W = P (V_2 - V_1)$

C) Isothermal Process: (constant temperature process)

Temperature remain constant ($T = \text{constant}$)

for ideal gas $= PV = \text{constant}$

for work done $= W = nRT \ln \left(\frac{V_2}{V_1} \right)$

D) Quasi-static Process:

A process that happen very slowly, so system remain in thermodynamic equilibrium at every stage. It is nearly reversible.

E) open system: mass does not cross boundary, but energy can.

ex: - piston-cylinder system.

F) closed system: mass does not crosses boundary

ex: - Ideal thermo flask

G) Isolated system: Neither mass nor energy crosses boundary ex:- Ideal thermos flask.

H) System: A defined quantity of matter or region in space selected for study.

I) Boundary: Surface separating system from surroundings can be real or imaginary.

J) Surrounding: Everything outside the system.

K) Process: Change of state of a system.

L) Cyclic process: System returns to initial state
 $\Delta U = 0$

M) Point Function: Depends only on state
ex:- pressure, Temperature, volume, Internal energy.

N) Path function: Depends on mass
example: mass, volume, energy.

O) Extensive property: Depends on mass
example: ~~Temperature, Pressure~~
mass, volume, energy

P) Intensive property: Independent of mass
ex:- ~~Independent of mass~~ Temp, pressure

Q) PNM - 1 (Perpetual motion machine of First kind)
machine that produces work without energy input. (Violates First law of Thermodynamics)

Impossible

R) First law of energy

S) First

since

T) Limit

i) does

ii) does

iii) does

U) micro mole

deals

V) macros study

W) Role in Therm

. energy

. En

. No

X) zeroth

If A

with C

with O

R) First law of Thermodynamics (For Process):
energy can neither be created nor destroyed.

$$q - w = \Delta u$$

S) First law of Cycle Process:

$$\oint \delta q = \oint \delta w$$

Since

$$\Delta u = 0$$

T) Limitations of First law:

- i) does not give direction of heat flow.
- ii) does not explain irreversibility.
- iii) does not give efficiency limit.

U) microscopic point of view: Study based on molecular behaviour

deals with kinetic theory.

V) macroscopic point of view:-

Study based on measurable like P, V, T

W) Role in Quantum Thermodynamics:

Thermodynamic applied to quantum system.

- energy quantization
- Entropy in microscopic system.
- Nano-scale heat engines

X) Zeroth law of Thermodynamics.

If A and B are in thermal equilibrium with C, then A and B are in equilibrium with each other.

Y) Thermodynamic Equilibrium:

System must satisfy:

- 1) Thermal equilibrium
- 2) mechanical equilibrium
- 3) chemical equilibrium

Z) Role of Thermodynamic in Daily life

- 1) Refrigerators
- 2) Air conditioners
- 3) Automobiles
- 4) Power plants
- 5) Human body metabolism
- 6) cooking

Question-2)

$$P V^\gamma = c$$

$$W = \int P dV$$

$$P = \frac{c}{V^\gamma}$$

$$W = \int \frac{c}{V^\gamma} dV$$

$$W = \frac{c}{1-\gamma} (V_2^{1-\gamma} - V_1^{1-\gamma})$$

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$

question-3)

$$P V^n = \text{constant}$$

work done

$$W = \frac{P_2 V_2 - P_1 V_1}{1-n}$$

question-4)

a) steady flow energy equation (SFEE)
for open system

$$\dot{Q} - \dot{W} = (h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$$

for jet engine (steady state)

$$\dot{Q} - \dot{W} = \dot{m} \left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right]$$

b) Assumptions:

- i) steady state
- ii) no accumulation of mass
- iii) one inlet, one outlet
- iv) negligible potential energy change

ques-5)

- sum of heat transfer per cycle = 3000 KJ
- cycle speed = 600 cycle/hrs
- complete cycle

$$\Sigma \Delta U = 0$$

First law:

$$Q - W = \Delta U$$

for complete cycle

$$\Sigma Q = \Sigma W$$

$$W_{\text{cycle}} = 3000 \text{ KJ}$$

ques-6)

- Net heat cycle = 30 KJ
- 1000 cycle/min

$$W_{\text{cycle}} = 30 \text{ KJ}$$

• work rate

$$\begin{aligned} W_{\text{rate}} &= 30 \times 1000 \\ &= 30000 \text{ KJ/min} \end{aligned}$$

ques-7)

- Heat per cycle = 50 KJ
- 100 cycle/sec

$$W_{\text{cycle}} = 50 \text{ KJ}$$

• work rate

$$W_{\text{rate}} = 50 \times 100 = 5000 \text{ KJ/sec}$$

question-8

$$T_m = [R (\ln F) + \mu + T_c]$$

Ice point

$$F_1 = 24.441$$

$$T_{mb} = 32^\circ F$$

Steam point

$$F_2 = 50$$

$$T_{mb} = 212^\circ F$$

Let :

$$T_m = a \ln F + b$$

using calibration

$$\text{At ice point} = 32 = a \ln (24.441) + b$$

$$\text{At steam point} = 212 = a \ln (50) + b$$

Solve simultaneously :

$$\ln (24.441) \approx 3.197$$

$$\ln (50) \approx 3.912$$

$$\text{Equation 1: } 32 = 3.197a + b$$

$$\text{Equation 2: } 212 = 3.912a + b$$

$$\text{Subtract : } 180 = 0.715a$$

$$a \approx 251.75$$

Now substitute -

$$32 = 3.197 (251.75) + b$$

$$32 = 804.8 + b$$

$$b = -772.8$$

Now for $F = 36.80$

$$\ln (36.80) \approx 3.605$$

$$T_m = 251.75 (3.605) - 772.8$$

$$T_m \approx 906.8 - 772.8$$

$$T_m \approx 134^\circ F$$

ques - 9)

Ice point = 250°S

Steam point = 500°S

F values

25 and 50

use linear relation

$$T_S = aF + b$$

$$At\ F = 25 \rightarrow T = 250$$

$$At\ F = 50 \rightarrow T = 500$$

Solve:

$$250 = 25a + b$$

$$500 = 50a + b$$

subtract

$$250 = 25a$$

$$a = 10$$

$$\text{Now: } 250 = 25(10) + b$$

$$250 = 250 + b$$

$$b = 0$$

So

$$T_S = 10F$$

$$\text{for } F = 36.80$$

$$T_S = 10 \times 36.80$$

$$= 368^{\circ}\text{S}$$

ques - 10)

$$T = n (\ln \phi) + \lambda$$

Ice point (0°C): $\phi_1 = 51$

Steam point (100°C): $\phi_2 = 100$

Let

$$T = a \ln \phi + b$$

$$\text{At } \phi = 51 \rightarrow T = 0$$

$$0 = a \ln(51) + b$$

$$\text{At } \phi = 100 \rightarrow T = 100$$

$$100 = a \ln(100) + b$$

$$\ln(51) \approx 3.932$$

$$\ln(100) = 4.605$$

Equation 1:

$$0 = 3.932a + b$$

Equation 2:

$$100 = 4.605a + b$$

Subtract:

$$100 = 0.673a$$

Now:

$$0 = 3.932(148.6) + b$$

$$b = -584.4$$

$$\text{For } \phi = 138$$

$$\ln(138) \approx 4.927$$

$$T = 148.6(4.927) - 584.4$$

$$T \approx 732 - 584.4$$

$$T = 148^\circ\text{C}$$

Ques-11)

$$P_1 = 3 \text{ bar}$$

$$V_1 = 3 \text{ m}^3$$

$$P_2 = 1.2 \text{ bar}$$

$$\text{Law: } PV^2 = C$$

Step 1) Find constant C

$$C_1 = P_1 V_1^2$$

$$C = 3 \times 3^2$$

$$C = 3 \times 9 = 27$$

$$\text{So, } PV^2 = 27$$

Step 2) Find V_2

$$P_2 V_2^2 = 27$$

$$1.2 V_2^2 = 27$$

$$V_2^2 = 22.5$$

$$V_2 = 4.743 \text{ m}^3$$

Step 3) Work done

$$PV^n = C$$

$$W = \frac{P_2 V_2 - P_1 V_1}{1-n} \quad \because n=2$$

$$W = \frac{(1.2 \times 4.743) - (3 \times 3)}{1-2}$$

$$W = 5.694 \text{ kJ}$$

$$W = 331 \text{ kJ}$$

$$W = 3.311 \text{ bar} \cdot \text{m}^3$$

ques-12)

$$\bullet P_1 = 25 \times 10^5, P_2$$

$$\bullet V_1 = 36 \text{ m}^3$$

$$\bullet V_2 = 100 \text{ m}^3$$

$$\bullet 2 \text{ a w } P = C/V^2$$

$$\rightarrow C = P_1 V_1^2$$

$$C = 25 \times 10^5 \times 36^2$$

$$~~36^2 = 1296~~$$

$$C = 3.24 \times 10^{10}$$

$$\rightarrow W = \int P dV$$

$$W = \int \frac{C}{V^2} dV$$

$$W = C \int V^{-2} dV$$

$$W = C \left(\frac{1}{V_1} - \frac{1}{V_2} \right)$$

$$W = 3.24 \times 10^{10} \left(\frac{1}{36} - \frac{1}{100} \right)$$

$$W = 3.24 \times 10^{10} (0.02778 - 0.01)$$

$$W = 3.24 \times 10^{10} (0.01778)$$

$$W \approx 5.76 \times 10^8 \text{ J}$$

$$W \approx 576 \text{ MJ}$$

ques-13)

$$PV = 25$$

$$P_1 = 5 \text{ bar}, \quad P_2 = 10 \text{ bar}$$

• step-1 : Find vol

$$V = \frac{25}{P}$$

$$V_1 = 25/5 = 5$$

$$V_2 = 25/10 = 2.5$$

• step-2 : work done

$$PV = \text{constant}$$

Isothermal type

$$W = P_1 V_1 \ln(V_2/V_1)$$

$$W = 25 \ln(2.5/5)$$

$$W = 25 \ln(0.5)$$

$$\ln(0.5) = -0.693$$

$$W = -17.33 \times 10^3$$

$$W = -1733 \text{ KJ}$$

8-14)

$$PV = 15$$

$$P_1 = 9.25 \times 10^5 \text{ Pa}$$

$$P_2 = 3.40285 \times 10^5 \text{ Pa}$$

Step-① Convert to bar

$$1 \text{ bar} = 10^5 \text{ Pa}$$

$$P_1 = 9.25 \text{ bar}$$

$$P_2 = 3.4025 \text{ bar}$$

Step-② volume

$$V = 15/P$$

$$V_1 = 15/9.25 = 1.62$$

$$V_2 = 15/3.40285 = 4.41$$

Step-③

$$W = 15 \ln \left(\frac{4.41}{1.62} \right)$$

$$= 15 \ln (2.72)$$

$$W = 15 \times 10^5$$

$$W \approx 1500 \text{ kJ}$$

8-15)

$$\Delta u = 0$$

$$W_{net} = Q_{net}$$

All Q values from table

$$Q_{net} = 0$$

So,

$$W_{net}$$

$$\text{Net work done} = 0 \text{ kJ}$$

ques-16)

for cyclic process.

$$\Delta u = 0$$

from first law

$$q_{net} = w_{net}$$

$$w_{net} = 0$$

so

$$q_{net} = 0$$

$$\text{Net heat transfer} = 0 \text{ kJ}$$

$$\text{Net internal energy change} = 0$$